

## EDUCATION

**The Graduate Center, City University of New York (CUNY)** — New York, NY

Ph.D. Candidate in Physics, *expected 2027*

Research Focus: Quantum sensing with NV center qubits, spin control, and optical spectroscopy for nanoscale metrology and fluid dynamics.

**The City College of New York, CUNY** — New York, NY

M.Phil. in Physics, 2025

**Brooklyn College, CUNY** — Brooklyn, NY

B.S. in Physics, magna cum laude, May 2020

Minors: Computer Science and Mathematics | GPA: 3.8

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## RESEARCH EXPERIENCE

**Meriles Group, CUNY City College** — New York, NY

Researcher, 2022–Present

- Designed and optimized quantum sensing experiments using NV center qubits for nanoscale NMR to study nanoscale water dynamics under confinement.
- Developed optical and microwave control sequences for spin initialization, readout, and coherence optimization.
- Calibrated confocal optics and fluorescence detection systems for single-photon measurements.
- Co-author on publications exploring fluid behavior at the nanoscale and NV-based sensing techniques.
- Mentored two high school students in lab techniques, simulations, and data acquisition as part of outreach mentorship initiatives.

**Materials Research Laboratory Prof. Sophia Suarez, Brooklyn College, CUNY** — Brooklyn, NY

Research Assistant, *Summer 2020, Summer 2021*

- $^1\text{H}$  NMR studies on surfactant-enhanced gas hydrate formation and growth.

**RAMREU in Combinatorics and Graph Theory, CUNY** — Summer 2019

- Applied Python for complex network analysis and combinatorial data visualization.
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## TEACHING EXPERIENCE

**Adjunct Lecturer – City College of New York (CUNY)**

*Jan 2022 – Present*

- Instructor for Physics I & II Lab courses (20+ students/semester).
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## SKILLS

**Quantum & Experimental Hardware:** NV center qubits, quantum sensing, magnetic resonance, NMR spectroscopy, confocal optics, fluorescence microscopy, vacuum systems, FPGA

**Electronics & Fabrication:** Circuit design, soldering, signal conditioning, wiring, 3D printing, mechanical design (Autodesk Inventor), electronics repair and assembly

**Programming:** Python, MATLAB, Java, LaTeX

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## HONORS & AWARDS

- CREST IDEALS Fellowship (2023–2025)
  - 1st Place, *Global NMR Conference* (2025)
  - 2nd Place, *Quantum Creators Con* (QuEra, 2025)
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## LEADERSHIP & ENGAGEMENT

**Organizer, APS CUWiP at CUNY** (2023) — Coordinated hotel rooms, catering, applications, and website; secured invited speakers and managed logistics.

**Student Representative, DGSC – Physics Program, CUNY Graduate Center** (2024–Present) — Represented physics graduate students in departmental and council meetings.